

Activation Energy (E_a) & Reaction Energy Diagrams

Activation energy • bond energy changes • reaction profiles

🌱 Big Idea

Activation energy is the “start-up cost” of a chemical change. Even when a reaction releases energy overall (exothermic), the reacting particles still need an initial push to reach a high-energy arrangement called the transition state. Once particles make it over that energy barrier, new bonds can form and the reaction can proceed.

If you remember only one idea from today, make it this: a reaction can be “allowed” by energy (ΔH) and still not occur unless particles can climb the E_a hill.

🔑 1) Activation energy: why reactions need a “push”

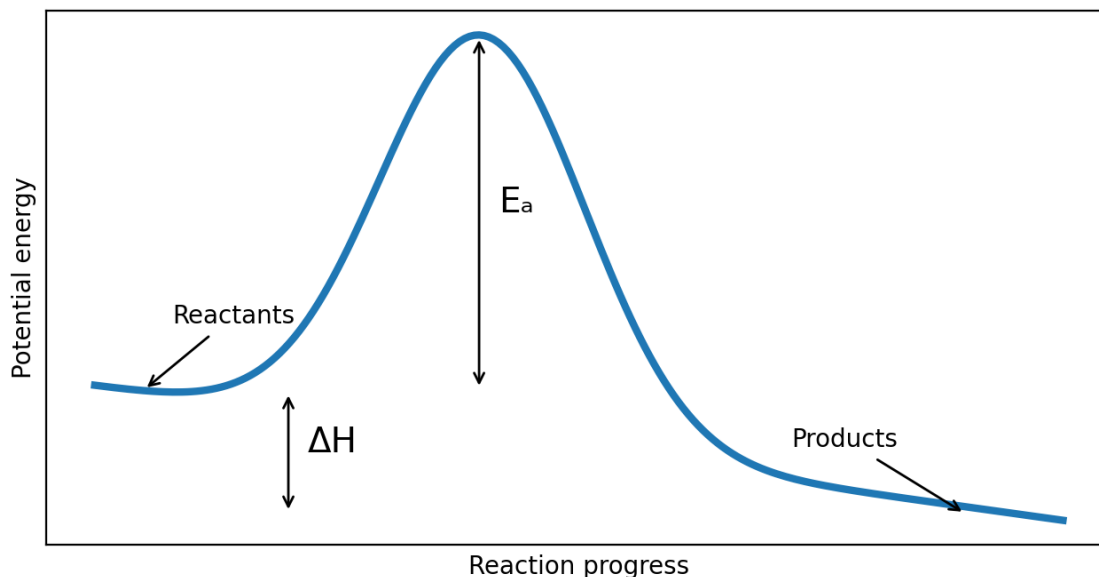
A common misconception is to think that “exothermic” means “will happen automatically.” In chemistry, those are not the same thing. Exothermic tells you about the energy difference between products and reactants (ΔH), but it says nothing about how hard it is to start the change. Activation energy (E_a) is the reason some reactions are energetically favourable yet still slow—or appear not to happen at all—at room temperature.

To understand why E_a exists, picture what must happen at the particle level. Reactant particles are held together by attractive forces and chemical bonds. For a reaction to occur, particles have to collide, approach closely, and begin rearranging electrons. That rearrangement usually requires some existing bonds to stretch and partially break before new bonds can form. Stretching and breaking bonds requires energy input, so the system’s potential energy rises temporarily during the “getting started” stage.

That temporary rise is the energy barrier. The peak of the barrier corresponds to the transition state (sometimes called the activated complex). At this point, the arrangement of atoms is unstable: old bonds are weakened, new bonds are only partly formed, and the system is at its

highest potential energy along the reaction path. Once the transition state is reached, the system can move downhill as stronger, more stable bonds form.

Another way to say the same idea is: activation energy controls reaction rate because only collisions with enough kinetic energy (and the right orientation) can produce the transition state. At any temperature, particles have a range of kinetic energies. Raising temperature increases the fraction of particles that have enough energy to get over the barrier, which is why heating a system often speeds reactions dramatically.



Notice two different vertical “distances” on the diagram. The height from reactants up to the peak is the activation energy (E_a). The difference between reactants and products is the enthalpy change (ΔH). These are related but they are not interchangeable. E_a is about the barrier and therefore the rate; ΔH is about where the system starts and ends and therefore the net energy transfer.

Key Vocabulary

Activation energy (E_a): the minimum energy required for reactant particles to reach the transition state.

Transition state (activated complex): the unstable, high-energy arrangement of atoms at the top of the energy barrier.

Reaction profile / energy diagram: a graph of potential energy vs. reaction progress that shows E_a and ΔH .

ΔH (enthalpy change): the net energy change of the system from reactants to products at constant pressure.

Common Error

Mixing up E_a and ΔH . If a graph is provided, E_a is measured from the reactants (or products, for the reverse reaction) up to the peak. ΔH is measured from reactants down/up to products. A catalyst changes E_a but does not change ΔH .

Diploma Tip

When you are asked to label an energy diagram, label the axes first: potential energy (y) and reaction progress (x). Then label reactants, products, the peak (transition state), E_a (forward and/or reverse), and ΔH . Remember: E_a values are always positive numbers, while ΔH can be positive or negative depending on whether the reaction is endothermic or exothermic.

Pause & Think

Gasoline and oxygen can coexist for months in a sealed container, yet the combustion reaction is highly exothermic. Explain, using activation energy, why the reaction does not occur until a spark is provided.

Answer (self-check): without a spark, too few particle collisions have enough energy to reach the transition state. The spark provides the initial energy needed to overcome E_a , allowing the first successful collisions to occur. Once combustion starts, the heat released helps keep supplying energy to new collisions, so the reaction becomes self-sustaining.

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Activation Energy (E_a) — Part B

Bond breaking vs bond forming • bond energies • the “investment then payoff” idea • connecting E_a and ΔH

Big Idea

A reaction’s energy change can feel confusing at first because two things are always happening at once: bonds in the reactants are being broken (which requires energy), and new bonds in the products are being formed (which releases energy).

The overall enthalpy change, ΔH , is the net result of those two competing energy changes. Activation energy exists because the bond-breaking part comes first — the reaction must “pay” energy up front to reach a high-energy arrangement (the transition state) before it can “collect” energy back as new bonds form.

Part B — Bond Breaking & Bond Forming: Where Does the Energy Go?

When students first hear “energy is stored in bonds,” it often sounds backwards. If a bond is “energy,” shouldn’t breaking it release that energy? The truth is the opposite, and getting this straight early makes the rest of thermochemistry much easier.

A chemical bond forms because the bonded arrangement has a lower potential energy than the separated atoms. Nature tends to move toward lower potential energy states. When a bond forms, energy is released to the surroundings as the system drops into a more stable (lower-energy) arrangement.

Breaking a bond means pulling atoms out of that low-energy arrangement. That requires an input of energy to overcome the attractive forces holding the atoms together. So bond breaking is always an energy-absorbing step (endothermic for the system). Bond forming is always an energy-releasing step (exothermic for the system).

One useful way to keep this straight is to compare the two processes side-by-side:

Process	Energy story (system view)	Picture it like...
Breaking bonds	Energy must be absorbed to pull particles apart. Potential energy increases.	Pushing up a hill / stretching a spring
Forming bonds	Energy is released as particles settle into a lower-energy arrangement. Potential energy decreases.	Rolling down a hill / snapping a spring back

Because both bond breaking and bond forming occur in the same reaction, the overall enthalpy change is the difference between the energy absorbed to break bonds and the energy released when new bonds form:

$$\Delta H \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$$

Notice two important things here. First, the symbol “ $\approx$ ” reminds you that bond energies are average values, so this method produces an estimate. Second, the sign of ΔH depends on which side “wins.” If forming bonds releases more energy than breaking bonds absorbs, ΔH is negative (exothermic). If breaking bonds requires more energy than forming bonds releases, ΔH is positive (endothermic).

Now connect this back to activation energy. At the start of a reaction, bonds in the reactants must be stretched and partially disrupted before new bonds can form. That initial stretching and breaking is the “uphill” energy investment. The highest-energy arrangement along the way is called the transition state (sometimes called the activated complex). The activation energy, E_a , is the energy required to reach that point from the reactants.

This explains a common real-world observation: gasoline combustion is strongly exothermic, yet gasoline does not spontaneously burn in a tank. The reaction is energetically favourable ($\Delta H < 0$), but it still requires an ignition source to supply E_a . Thermochemistry tells you whether a reaction releases or absorbs energy overall; activation energy helps explain whether it starts easily.

Worked Example 1 — Estimating ΔH using bond energies

Estimate the enthalpy change for: $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$

Step 1: Identify bonds broken (reactants). You must break one H-H bond and one Cl-Cl bond.

Step 2: Identify bonds formed (products). You form two H-Cl bonds.

Step 3: Use bond energies (typical values): H-H = 436 kJ/mol, Cl-Cl = 242 kJ/mol, H-Cl = 431 kJ/mol.

Energy absorbed (broken) = $436 + 242 = 678$ kJ

Energy released (formed) = $2(431) = 862$ kJ

$\Delta H \approx 678 - 862 = -184$ kJ

Interpretation: ΔH is negative, so the reaction is exothermic. The key reason is that the H-Cl bonds formed are collectively stronger (more energy released) than the H-H and Cl-Cl bonds that had to be broken.

Worked Example 2 — Exothermic does not mean “no energy in”

Estimate ΔH for: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$

Bonds broken: two H-H bonds and one O=O bond. Bonds formed: four O-H bonds.

Typical bond energies: H-H = 436 kJ/mol, O=O = 498 kJ/mol, O-H = 463 kJ/mol.

Energy absorbed (broken) = $2(436) + 498 = 1370$ kJ

Energy released (formed) = $4(463) = 1852$ kJ

$\Delta H \approx 1370 - 1852 = -482$ kJ

Even though the net change is very exothermic, the reaction still requires an initial input of energy to start breaking bonds and reach the transition state. That “start-up cost” is activation energy, and it is shown as the hump on an energy diagram.

Quick Self-Check: If a reaction has $\Delta H > 0$, what must be true about the balance of bond energies? Explain in one sentence.

Hint: Compare the total energy needed for bonds broken with the total energy released when bonds form.

Key Vocabulary

Bond energy (bond dissociation energy): the energy required to break one mole of a specific bond in the gas phase (reported as an average value).

Transition state (activated complex): a short-lived, high-potential-energy arrangement where old bonds are partly broken and new bonds are partly formed.

Activation energy (E_a): the minimum energy needed for reactants to reach the transition state and begin converting into products.

Enthalpy change (ΔH): the net energy change of the system for a reaction at constant pressure; it compares the energy of reactants and products, not the height of the barrier between them.

Common Error

“Breaking bonds releases energy.”

This is one of the most common misconceptions in Chemistry 30. Breaking a bond always requires energy input because you are separating particles that are attracted to one another. The energy released in an exothermic reaction comes from forming new, stronger bonds in the products — not from breaking the old ones.

Another common slip is mixing up ΔH and E_a . ΔH compares reactants to products (the start and end). E_a describes the energy barrier (the hump) that must be overcome on the way.

Diploma Tip

On written-response questions, diploma markers reward clear cause-and-effect language. If you are asked to justify why an exothermic reaction still needs activation energy, explicitly mention that bonds must be stretched and broken first (energy absorbed) to reach a transition state, and only then can new bonds form (energy released).

If you are using bond energies to estimate ΔH , always show: (1) bonds broken, (2) bonds formed, (3) correct coefficient multiplication. A sign error is usually a setup error — re-read the formula and ask yourself which side is “paying” energy and which side is “getting” energy back.

Pause & Think

1) Gasoline combustion has $\Delta H < 0$, yet gasoline can sit for months without reacting. In your own words, explain how activation energy resolves this “contradiction.”

2) A student says: “If a reaction is exothermic, it must happen quickly because it releases energy.” Explain why this reasoning is incomplete. Use the ideas of ΔH and E_a in your explanation.

Optional challenge: Find one real-life example of a reaction that is energetically favourable but slow at room temperature (rusting is a classic example). What provides the activation energy in your example?

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Lesson 6: Activation Energy

Part C: Energy Diagrams & Reaction Profiles

Big Idea

Energy diagrams are the “map” that separates two different ideas: **the energy barrier (activation energy)** and **the overall energy change (ΔH)**.

If you can read and label a reaction profile correctly, you can explain why a reaction might be exothermic but still slow, and you can predict what a catalyst changes (and what it does not).

In this part, we will learn to interpret energy diagrams the way diploma questions expect: by identifying reactants/products, locating the activated complex, and calculating E_a and ΔH from the diagram.

Part C — Energy Diagrams & Reaction Profiles

Process	Energy story (system view)	Picture it like...
Breaking bonds	Energy must be absorbed to pull particles apart. Potential energy increases.	Pushing up a hill / stretching a spring
Forming bonds	Energy is released as particles settle into a lower-energy arrangement. Potential energy decreases.	Rolling down a hill / snapping a spring back

1) Start with the question: what kind of “energy” is being graphed?

On an energy diagram, the vertical axis is labelled “potential energy” (sometimes “enthalpy” or simply “energy”). This is not the temperature of the reaction, and it is not the speed of particles. Instead, it represents the stored energy in the system due to how particles are arranged and how strongly they attract each other—especially through chemical bonds and intermolecular forces.

You can think of potential energy as the “stored” energy of the chemical system. When bonds form, energy is released and the system tends to move to a lower potential energy. When bonds break, energy must be absorbed and the system climbs to a higher potential energy. This is exactly the bond-breaking/bond-forming story from Part B, but now we are going to picture it.

2) What does the horizontal axis mean?

The horizontal axis is usually called “reaction progress” or “reaction coordinate.” This axis does not represent time. It also does not represent distance. It is a bookkeeping path that takes you from reactants to products. The point of this axis is simply to show the energy of the system as the reaction proceeds from “start” (reactants) to “finish” (products).

This is an important misconception to catch early: a reaction can move quickly or slowly along the reaction progress axis. The diagram is not telling you the rate directly. It is telling you the energy landscape the reacting particles must travel through.

3) Reactants, products, and the activated complex

Most reaction profiles have three key landmarks. First, you have the reactants (left side). Second, you have the products (right side). In between, you have a peak—the highest point on the curve. That peak represents the activated complex (also called the transition state).

The activated complex is not a stable substance you can collect in a jar. It is a momentary arrangement where old bonds are partly broken and new bonds are partly forming. Because it contains “half-broken” and “half-formed” bonds, it is an extremely high-energy arrangement. That is why the peak is higher than both the reactants and products.

4) Activation energy (E_a): the barrier that must be crossed

Activation energy is the minimum energy that reacting particles must have to reach the activated complex. On an energy diagram, activation energy is measured vertically from the reactant energy level up to the peak.

A key idea (and one the diploma exam loves to probe) is that even an exothermic reaction usually needs an activation energy. Exothermic means “products are lower than reactants,” not “no barrier.” A campfire releases lots of energy, but you still need a match. Gasoline can release enormous energy, but it still needs a spark. In both cases, the system has to be pushed up the hill before it can roll down.

5) Enthalpy change (ΔH): the net energy difference

ΔH is the overall change in energy from reactants to products. On the diagram, ΔH is measured vertically between the reactant level and the product level.

If products are lower than reactants, ΔH is negative and the reaction is exothermic (energy is released to the surroundings). If products are higher than reactants, ΔH is positive and the reaction is endothermic (energy is absorbed from the surroundings).

Notice that ΔH and E_a measure different things. E_a tells you how hard it is to get the reaction started. ΔH tells you where the energy ends up after the reaction is complete. A reaction can have a small ΔH but a large E_a , or a large ΔH but a small E_a . That difference matters when you explain reaction rate versus energy change.

6) Forward vs reverse activation energy

Every energy diagram contains two possible activation energies: one for the forward reaction (reactants $\rightarrow$ products) and one for the reverse reaction (products $\rightarrow$ reactants). The peak is the same in both directions, because the activated complex is the same arrangement of atoms.

Forward E_a is measured from reactants to the peak. Reverse E_a is measured from products to the peak. This means that if a forward reaction is exothermic (products lower), the reverse

reaction will have a larger activation energy—because it has to climb from a lower starting point up to the same peak.

Relationship to remember: $E_a(\text{reverse}) = E_a(\text{forward}) + |\Delta H|$ (for an exothermic forward reaction)

For an endothermic forward reaction, the products are higher. That means the reverse activation energy is smaller than the forward activation energy by exactly the magnitude of ΔH .

7) A picture to lock it in

The diagram below shows both an exothermic and endothermic profile. Focus on what the arrows measure: E_a goes to the peak, ΔH goes from reactants to products.

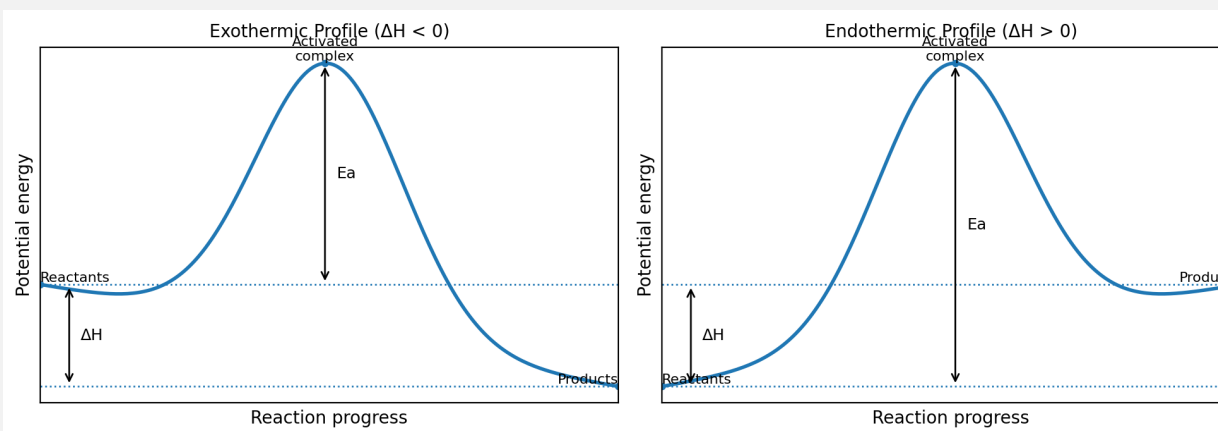


Figure: Reaction profiles showing E_a and ΔH (exothermic vs endothermic).

8) Worked Example 1 — Reading E_a and ΔH from energy values

An energy diagram shows reactants at 120 kJ, the peak at 170 kJ, and products at 80 kJ.

Step 1: Calculate forward activation energy.

$E_a(\text{forward}) = 170 \text{ kJ} - 120 \text{ kJ} = 50 \text{ kJ}$. This tells you the minimum “push” needed to reach the activated complex.

Step 2: Calculate ΔH .

$\Delta H = 80 \text{ kJ} - 120 \text{ kJ} = -40 \text{ kJ}$. The negative sign tells you the system loses energy overall (exothermic).

Step 3: Calculate reverse activation energy.

$E_a(\text{reverse}) = 170 \text{ kJ} - 80 \text{ kJ} = 90 \text{ kJ}$. Notice it is larger because the reverse reaction starts from the lower product level.

9) Worked Example 2 — Endothermic profile (do not confuse E_a and ΔH)

An energy diagram shows reactants at 50 kJ, the peak at 140 kJ, and products at 110 kJ.

$E_a(\text{forward}) = 140 - 50 = 90 \text{ kJ}$. $\Delta H = 110 - 50 = +60 \text{ kJ}$ (endothermic). $E_a(\text{reverse}) = 140 - 110 = 30 \text{ kJ}$. The reaction is endothermic overall, but it still has an activation barrier in both directions.

10) Worked Example 3 — Sketching a diagram from numbers

Suppose a forward reaction has $E_a = 70 \text{ kJ}$ and $\Delta H = -20 \text{ kJ}$.

Start by deciding exothermic or endothermic: ΔH is negative, so products must be drawn lower than reactants. Next, place a peak 70 kJ above the reactant level. Finally, place the product level 20 kJ below the reactant level. When you label the diagram, make sure your E_a arrow goes from reactants up to the peak, and your ΔH arrow goes from reactants down to products.

11) Quick Check — Try this before you scroll past

A reaction profile shows reactants at 80 kJ, the peak at 150 kJ, and products at 100 kJ.

a) Is the forward reaction exothermic or endothermic? b) Calculate $E_a(\text{forward})$ and ΔH .

Answer: $E_a = 150 - 80 = 70 \text{ kJ}$. $\Delta H = 100 - 80 = -20 \text{ kJ}$, so the forward reaction is exothermic.

12) Why this matters in Thermochemistry (Unit II)

In Unit II, you often calculate ΔH using calorimetry, Hess's Law, or formation enthalpies. Those methods tell you the net energy change between reactants and products. Energy diagrams help you explain that net change visually, and they help you keep ΔH separate from E_a .

This separation is crucial when you start discussing catalysts. A catalyst lowers E_a (the hill gets smaller), which can make a reaction faster, but it does not change ΔH . The reactant and product energy levels stay the same—only the pathway changes.



Key Vocabulary

Potential energy: stored energy due to particle arrangement and attractions (especially bonds).

Reaction progress (reaction coordinate): an axis that tracks the reaction from reactants to products; it is not time.

Activated complex (transition state): the highest-energy, unstable arrangement where bonds are partly broken/formed.

Activation energy, E_a : the energy barrier from reactants up to the activated complex (forward) or from products up to the activated complex (reverse).

ΔH (enthalpy change): the net energy difference between products and reactants; negative = exothermic, positive = endothermic.

Reaction profile / energy diagram: a graph of potential energy versus reaction progress used to visualize E_a and ΔH .

Common Error

Mixing up E_a and ΔH : E_a goes to the peak; ΔH goes from reactants to products. If you draw the wrong arrow, every calculation after that will be wrong.

Assuming exothermic means “easy”: exothermic reactions still have an activation barrier. Exothermic describes the energy difference between start and finish, not the hill in the middle.

Treating the x-axis as time: the diagram does not automatically tell you a reaction is fast or slow. The shape tells you about the energy barrier, and that barrier influences rate, but the axis itself is just a progress path.

Forgetting the reverse reaction: diploma questions often ask for $E_a(\text{reverse})$. Always measure from the starting level (reactants or products) up to the same peak.

Diploma Tip

When a diploma question gives you an energy diagram, the fastest way to avoid mistakes is to label first, calculate second. Write “R” over the reactant plateau, “P” over the product plateau, and circle the peak as the activated complex.

Then draw two vertical arrows: one from R up to the peak (E_a forward) and one from R down/up to P (ΔH). Only after the arrows are on the diagram should you plug numbers into equations.

If you need $E_a(\text{reverse})$, draw the arrow from P to the peak. Many students try to “flip the sign” instead of re-measuring the vertical distance. Do not do that—activation energy is always a positive energy barrier.

Pause & Think

You are given a reaction profile with these energy levels: reactants = 95 kJ, peak = 160 kJ, products = 70 kJ.

1) Calculate $E_a(\text{forward})$. 2) Calculate ΔH . 3) Without doing any new arithmetic, decide whether $E_a(\text{reverse})$ is larger or smaller than $E_a(\text{forward})$ —and explain why.

Check your thinking: $E_a(\text{forward}) = 65 \text{ kJ}$. $\Delta H = -25 \text{ kJ}$ (exothermic). $E_a(\text{reverse})$ must be larger because the reverse reaction starts lower (products) but climbs to the same peak.

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Lesson 6: Activation Energy

Part D: Activation Energy vs ΔH

Big Idea

Activation energy (E_a) and enthalpy change (ΔH) are both “energy numbers,” but they answer two completely different questions.

E_a tells you how much energy the reactant particles must have to reach the activated complex (the top of the energy hill). ΔH tells you how much energy the system loses or gains overall between reactants and products.

If you can read an energy diagram and identify these two distances correctly, you will avoid the most common Thermochemistry/Kinetics mix-ups on tests and diploma-style questions.

Part D — Activation Energy vs ΔH

Mini-lesson 1 — Two different energy questions: When students look at an energy diagram, the most common mistake is assuming there is only one “energy change” to talk about. In reality, an energy diagram contains two different stories at the same time: (1) the energy barrier you must climb to get started, and (2) the net energy change from start to finish. Activation energy (E_a) is the barrier story. Enthalpy change (ΔH) is the net-change story.

A helpful analogy is hiking. Imagine the reactants are at the bottom of a valley and the products are at a different elevation. To get from the valley to the final location, you may have to climb a hill first. The height of that hill above where you started is like E_a . The difference between your starting elevation and your final elevation is like ΔH . Notice that these are not the same thing—and they can point in different directions.

This explains a famous chemistry fact: some reactions release a lot of energy (large negative ΔH) but still do not happen quickly at room temperature. They are energy-favourable, but they have a high activation energy barrier. A classic example is diamond converting to graphite: the change is energetically favourable, but it does not happen noticeably because the barrier is enormous.

Mini-lesson 2 — What activation energy really means: Activation energy is the minimum energy that reacting particles must have, at the moment they collide, to reach the activated complex (also called the transition state). You can think of the activated complex as the “top of the hill” on an energy diagram—bonds are partly broken and partly formed, so the arrangement is unstable and high-energy.

Why does this barrier exist? Because breaking bonds requires energy. Before new product bonds can form and release energy, the reactant bonds must first be stretched and weakened. Even if the overall reaction releases energy (exothermic), you still have to invest some energy up front to start the bond-rearrangement process. That initial energy investment is what E_a measures.

First, activation energy is always reported as a positive number because it represents a vertical distance on an energy diagram. Second, activation energy is a kinetics idea: it helps explain reaction rate. A reaction with a small barrier can proceed readily at a given temperature because more collisions have enough energy to be successful. A reaction with a large barrier proceeds slowly because only a small fraction of collisions reach the required energy.

Mini-lesson 3 — What ΔH tells you (and what it does not): Enthalpy change, ΔH , describes the net energy change of the system as reactants become products at constant pressure. On an energy diagram, ΔH is the difference between the potential energy of the products and the potential energy of the reactants.

If products are lower in potential energy than reactants, ΔH is negative and the reaction is exothermic: the system releases energy to the surroundings. If products are higher in potential energy than reactants, ΔH is positive and the reaction is endothermic: the system absorbs energy from the surroundings.

Here is the key comparison to keep repeating until it feels automatic:

ΔH is about where you start and where you finish.

E_a is about what you must climb over in the middle.

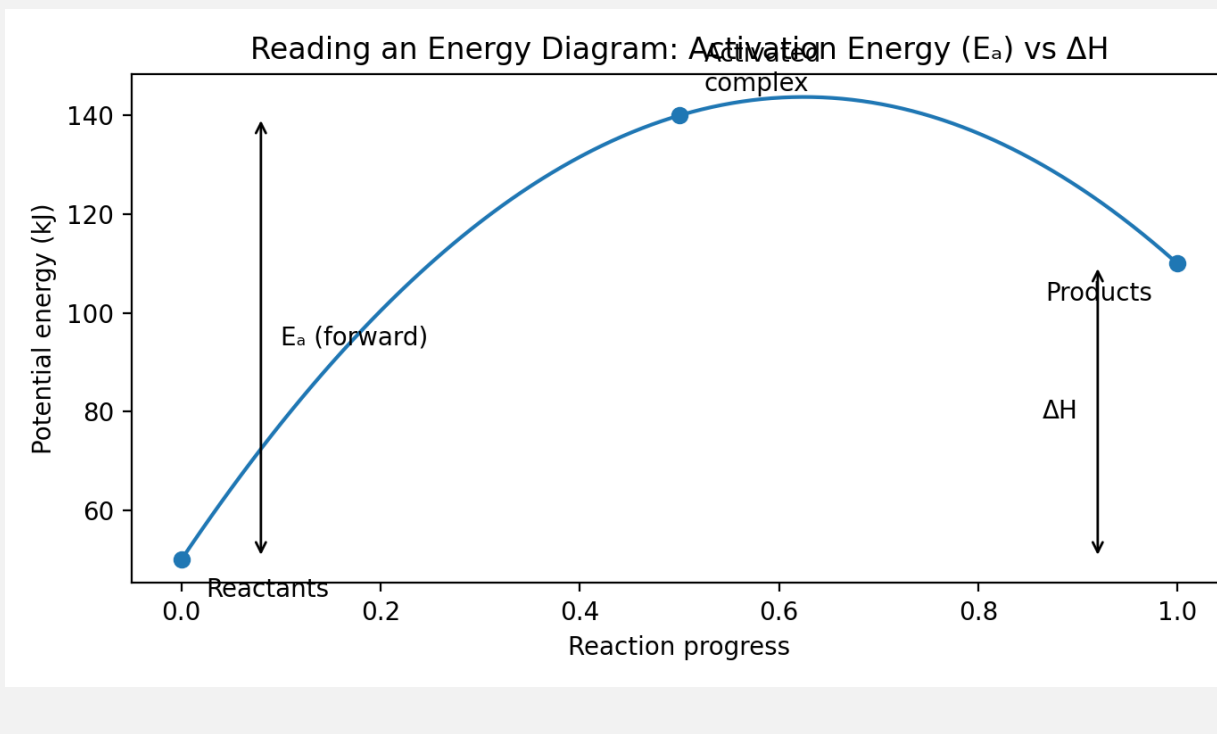
Mini-lesson 4 — Reading E_a and ΔH directly from an energy diagram: Most diploma-style energy-diagram questions are not trying to trick you with math. They are testing whether you can identify the correct vertical distances.

Step 1: Identify the three key energy levels. Find the reactants (left plateau), the activated complex (highest point on the curve), and the products (right plateau). Label them before you calculate anything. This prevents “wrong baseline” mistakes.

Step 2: Use the correct differences:

- E_a (forward) = $E(\text{activated complex}) - E(\text{reactants})$
- $\Delta H = E(\text{products}) - E(\text{reactants})$
- E_a (reverse) = $E(\text{activated complex}) - E(\text{products})$

Notice that ΔH never involves the peak. ΔH ignores the pathway and compares only initial and final states. That is why ΔH is called a state function.



Worked Example 1 — Exothermic profile (find E_a forward, ΔH , and E_a reverse)

An energy diagram shows reactants at 120 kJ, the activated complex at 200 kJ, and products at 80 kJ.

Start by finding E_a (forward). This is the climb from reactants to the peak: $E_a = 200 - 120 = 80$ kJ.

Now find ΔH . This compares products to reactants: $\Delta H = 80 - 120 = -40$ kJ. The negative sign tells you the system releases 40 kJ of energy per reaction as written (exothermic).

Finally, find E_a (reverse). For the reverse reaction, the starting point is the products, so the climb is from 80 kJ to 200 kJ: $E_a(\text{reverse}) = 200 - 80 = 120$ kJ.

Quick self-check: because the forward reaction is exothermic, the reverse barrier should be larger. Here, 120 kJ is larger than 80 kJ, which makes sense.

Worked Example 2 — Endothermic profile (why E_a and ΔH feel “opposite”)

An energy diagram shows reactants at 50 kJ, the activated complex at 140 kJ, and products at 110 kJ.

E_a (forward) is still the climb to the peak: $E_a = 140 - 50 = 90$ kJ.

ΔH compares products to reactants: $\Delta H = 110 - 50 = +60$ kJ. The positive sign means the system absorbs 60 kJ (endothermic).

E_a (reverse) uses the products as the starting point: $E_a(\text{reverse}) = 140 - 110 = 30$ kJ.

Notice the pattern: for an endothermic reaction, the products are higher than the reactants, so the reverse reaction starts closer to the peak. That is why $E_a(\text{reverse})$ is smaller.

Worked Example 3 — Using ΔH to connect forward and reverse activation energies

Suppose a reaction has $E_a(\text{forward}) = 60$ kJ and $\Delta H = -25$ kJ (exothermic). You are asked to determine $E_a(\text{reverse})$.

Picture the energy diagram: products must be 25 kJ lower than reactants. The peak is 60 kJ above the reactants. That means the peak is $60 + 25 = 85$ kJ above the products.

So $E_a(\text{reverse}) = 85$ kJ.

A useful relationship to remember (and to justify) is: $E_a(\text{reverse}) = E_a(\text{forward}) - \Delta H$. Because ΔH is negative for an exothermic reaction, subtracting a negative increases the value, giving the larger reverse barrier.

Worked Example 4 — Catalyst: what changes and what stays the same

A catalyst provides an alternative pathway with a lower peak energy. Imagine reactants are at 120 kJ and products are at 80 kJ (so $\Delta H = -40$ kJ). Without a catalyst, the peak is 200 kJ. With a catalyst, the peak becomes 160 kJ.

Without a catalyst: $E_a = 200 - 120 = 80$ kJ.

With a catalyst: $E_a(\text{catalyzed}) = 160 - 120 = 40$ kJ.

ΔH does not change because the reactants and products have not changed: $\Delta H = 80 - 120 = -40$ kJ in both cases.

This is a core diploma idea: catalysts change rate by lowering E_a , but they do not change the overall energy difference between reactants and products.

In Thermochemistry, ΔH is the quantity you will calculate most often because it tells you the energy transferred when a reaction occurs. However, energy diagrams frequently appear on assessments because they connect Thermochemistry (ΔH) to Kinetics (E_a). If you can confidently label both values, you are ready for multi-step pathway questions and for the transition into kinetics and equilibrium later in the course.



Key Vocabulary

Activation energy, E_a : The minimum energy required for reactant particles to reach the activated complex and react.

Activated complex (transition state): A short-lived, high-energy arrangement of atoms where bonds are partly broken and partly formed.

Reaction progress (reaction coordinate): A way to represent how far a reaction has proceeded from reactants to products; it is not time.

Potential energy diagram: A graph of potential energy vs reaction progress that shows reactants, products, the energy barrier, and ΔH .

Enthalpy change, ΔH : The net energy change of the system from reactants to products at constant pressure.

Catalyst: A substance that increases reaction rate by providing an alternative pathway with a lower activation energy; it is not consumed.

⚠ Common Error

⚠ **Mixing up the distances:** Students sometimes measure ΔH from the reactants to the peak. Remember: ΔH never uses the peak. ΔH compares products to reactants only.

⚠ **Forgetting that E_a is always positive:** E_a is a barrier height. Even for an exothermic reaction, E_a does not become negative.

⚠ **Switching the baseline when a reaction is reversed:** For the reverse reaction, the starting energy is the products. $E_a(\text{reverse})$ is measured from products up to the same activated complex.

⚠ **Assuming exothermic means “fast”:** A negative ΔH tells you energy is released overall, not how quickly the reaction happens. A large E_a can make an exothermic reaction slow at room temperature.

🎓 Diploma Tip

🎓 **Label first, calculate second:** On diploma-style items, the fastest students are not the ones who rush—they are the ones who label reactants, products, and the peak immediately. Once those three levels are labelled, E_a and ΔH become simple differences.

🎓 **Use the pattern to check yourself:** Exothermic reaction ($\Delta H < 0$): products are lower, so $E_a(\text{reverse})$ should be larger than $E_a(\text{forward})$. Endothermic reaction ($\Delta H > 0$): products are higher, so $E_a(\text{reverse})$ should be smaller than $E_a(\text{forward})$. If your numbers do not match the pattern, re-check which level you used as the baseline.

🎓 **Catalyst questions are high-value marks:** If a question mentions a catalyst, expect to state (and possibly justify) two facts: the activation energy decreases, and ΔH stays the same because the initial and final states do not change.

🛑 Pause & Think

Try this without looking back at the examples. The goal is to practise choosing the correct vertical distance and keeping your signs straight.

Quick Check:

An energy diagram shows reactants at 30 kJ, the activated complex at 90 kJ, and products at 70 kJ.

- 1) Is the reaction exothermic or endothermic?
- 2) Calculate E_a (forward).
- 3) Calculate ΔH .
- 4) Calculate E_a (reverse).
- 5) A catalyst lowers the peak from 90 kJ down to 80 kJ. Which answers change, and which stay the same?

Check your answers (after you try):

- 1) Endothermic, because the products (70 kJ) are higher than the reactants (30 kJ).
- 2) $E_a(\text{forward}) = 90 - 30 = 60 \text{ kJ}$.
- 3) $\Delta H = 70 - 30 = +40 \text{ kJ}$.
- 4) $E_a(\text{reverse}) = 90 - 70 = 20 \text{ kJ}$.
- 5) With the catalyst, only the activation energies change because only the peak changes. $E_a(\text{forward, catalyzed}) = 80 - 30 = 50 \text{ kJ}$ and $E_a(\text{reverse, catalyzed}) = 80 - 70 = 10 \text{ kJ}$. ΔH stays +40 kJ because reactants and products are unchanged.

 Chemistry 30 | Thermochemical Changes

Lesson 6: Activation Energy

Part E: Catalysts & Energy Diagrams

 Big Idea

A catalyst changes the pathway, not the payoff. It lowers the activation energy (E_a) by providing an alternative route to products, so more collisions have enough energy to react at the same temperature. A catalyst does not change ΔH , and it does not get used up.

Mini-lesson 1 — Why catalysts exist: Thermochemistry tells us whether a reaction releases or absorbs energy (ΔH), but it does not tell us whether the reaction will happen quickly. Many reactions are “energetically allowed” but still slow because particles must cross an energy barrier before bonds can rearrange. That barrier is the activation energy, E_a . If E_a is large, most collisions do not have enough energy to reach the transition state, so the reaction rate is low—even if ΔH is very negative.

A useful way to think about this is: ΔH tells you “where you end up,” while E_a tells you “how hard it is to get there.” Burning gasoline is strongly exothermic, but it does not ignite at room temperature unless you supply a spark, because the reacting molecules still need a push to reach the activated state. Catalysts exist because industry and biology often need reactions to occur quickly at reasonable temperatures and pressures.

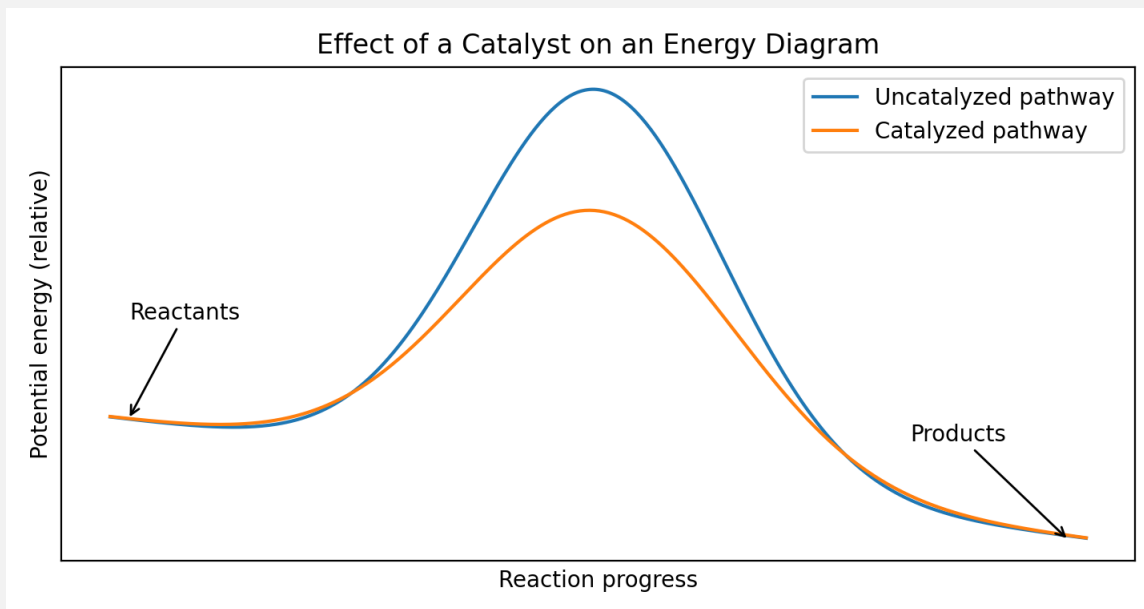
Mini-lesson 2 — What a catalyst actually does: A catalyst provides an alternative reaction pathway with a lower activation energy. This is not the same as “giving the reactants energy.” Instead, the catalyst changes the sequence of bond-breaking and bond-forming steps (the mechanism) so that the highest-energy point along the pathway is lower.

In practical terms, a catalyst often forms temporary interactions with reactants. Those interactions can weaken certain bonds, orient molecules correctly, or stabilize the transition state. Because the transition state is more stable, it requires less energy to reach. After the reaction, the catalyst is regenerated, which is why a small amount of catalyst can speed up a large amount of reaction.

A helpful analogy: lighting a log with a match is difficult because the log has a high “barrier” to ignition. If you add paper or kindling, the fire can start more easily and then spread to the log. In chemistry, a catalyst plays a similar role—it provides an easier route to the same products without changing the overall energy difference.

Mini-lesson 3 — Reading energy diagrams with and without a catalyst: On an energy diagram (potential energy vs reaction progress), a catalyst changes the height of the hump, not the starting and ending levels.

The reactant and product energy levels stay the same because they describe the chemical identities of the reactants and products. Since ΔH depends only on the difference between those two levels, ΔH does not change when a catalyst is used. What changes is E_a (the vertical distance from reactants up to the peak). For the reverse reaction, $E_a(\text{reverse})$ also decreases, because the catalyst provides an easier pathway in both directions.



Notice in the diagram that both curves begin at the same reactant level and end at the same product level. That is your visual proof that ΔH is unchanged. The catalyzed curve has a lower peak, which means a smaller E_a , and therefore a faster reaction at the same temperature.

Mini-lesson 4 — Multi-step pathways (why some catalyzed diagrams have two humps):

Some catalysts change a one-step pathway into a two-step pathway with an intermediate. In that case, the diagram may show two peaks. The rate is controlled by the highest peak (the rate-determining step), because that step has the largest E_a .

Even if a catalyzed pathway has two peaks, the overall ΔH is still found by comparing the reactant level to the product level. Intermediates affect the pathway, not the net energy difference between the starting and ending states.

Mini-lesson 5 — Catalysts, energy efficiency, and real systems: Catalysts matter because they allow reactions to run at lower temperatures or shorter times, which saves energy and reduces costs.

In cars, catalytic converters speed up redox reactions that convert CO and NO_x into less harmful gases before they leave the exhaust. In industry, catalysts make processes like ammonia production (fertilizers) practical at large scale. In biology, enzymes are catalysts that make life possible at body temperature—without enzymes, many essential reactions would be far too slow.

When you connect this to Thermochemistry, think about it this way: ΔH tells us the energy transferred, but catalysts make it possible to access that energy safely and efficiently. A reaction that releases a lot of energy is only useful if we can control the rate at which it releases that energy.

Worked Example 1 — Extracting E_a and ΔH from an energy diagram: Suppose the reactants are at 40 kJ, the peak is at 120 kJ, and the products are at 10 kJ (all relative values).

Forward activation energy: $E_a(\text{forward}) = 120 - 40 = 80$ kJ. Reverse activation energy: $E_a(\text{reverse}) = 120 - 10 = 110$ kJ. Enthalpy change: $\Delta H = 10 - 40 = -30$ kJ (exothermic).

If a catalyst provides a new pathway with a peak of 90 kJ, then $E_a(\text{forward})$ becomes $90 - 40 = 50$ kJ and $E_a(\text{reverse})$ becomes $90 - 10 = 80$ kJ. ΔH stays -30 kJ because the reactant and product levels did not move.

Worked Example 2 — Identifying the catalyzed pathway: Two diagrams show the same reactants and products. Diagram A has a peak at 150 kJ; Diagram B has a peak at 100 kJ. Which is catalyzed?

Diagram B is the catalyzed pathway because it has the lower peak (lower E_a). The correct reasoning is: catalysts reduce the barrier height, not the start/end energies.

Worked Example 3 — Catalyst vs temperature (conceptual): A student says, “A catalyst makes a reaction exothermic.” How do you respond?

You would point out that “exothermic” describes ΔH , which is determined by the energies of reactants and products. A catalyst does not change those energies, so it cannot change whether a reaction is exothermic or endothermic. What a catalyst changes is the pathway and therefore the activation energy. At the same temperature, lowering E_a means a larger fraction of particles have enough energy to react in collisions, so the reaction rate increases.

Temperature can also increase reaction rate, but it does so by raising particle kinetic energy overall. Catalysts increase reaction rate without increasing the average kinetic energy of particles—this is why catalysts are linked to energy efficiency.



Key Vocabulary

Activation energy (E_a): the minimum energy required for reacting particles to reach the transition state and form products.

Catalyst: a substance that increases reaction rate by lowering E_a through an alternative pathway; it is regenerated and not consumed overall.

Catalyzed pathway / alternative pathway: a different sequence of steps (mechanism) with a lower highest-energy point.

Transition state (activated complex): a short-lived, high-energy arrangement of atoms at the top of the energy barrier.

Heterogeneous vs homogeneous catalyst: heterogeneous catalysts are in a different phase from the reactants (often solid with gas/liquid reactants); homogeneous catalysts are in the same phase as the reactants (often all in solution).

Enzyme: a biological catalyst (usually a protein) that speeds up reactions in living systems by lowering E_a in a very specific way.

Common Error

Mistake: “A catalyst adds energy to the reactants.” Correction: A catalyst changes the pathway so less energy is needed to reach the transition state. It does not supply energy like heat or a spark.

Mistake: “A catalyst changes ΔH .” Correction: ΔH depends only on the energies of reactants and products. Catalysts do not change those levels, so ΔH stays the same.

Mistake: “A catalyst only speeds up the forward reaction.” Correction: A catalyst lowers the barrier in both directions, so both forward and reverse reactions speed up.

Mistake: “A catalyst shifts equilibrium toward products.” Correction: Catalysts do not change equilibrium position or K ; they help the system reach equilibrium faster by speeding both directions equally.

Diploma Tip

On diploma-style energy diagram questions, marks often come from correct labels and correct reasoning. When a catalyst is involved, always draw or identify TWO curves: uncatalyzed (higher peak) and catalyzed (lower peak).

Then label clearly: (1) E_a (forward) from reactants to the peak; (2) E_a (reverse) from products to the peak; and (3) ΔH from reactants down/up to products. If the reactant and product lines are unchanged, state explicitly that ΔH is unchanged.

If the question mentions equilibrium, include the key statement: “A catalyst does not change K or the equilibrium position; it only changes how quickly equilibrium is reached.” This is one of the most common high-value “concept marks” in Unit II.

Pause & Think

- 1) Sketch a reaction energy diagram for an exothermic reaction. Then add a second curve showing a catalyzed pathway. Label E_a (both pathways) and ΔH .
- 2) A reaction has $\Delta H = -250 \text{ kJ}$. Which statement is correct: (a) the catalyst makes ΔH less negative, (b) the catalyst makes ΔH more negative, or (c) the catalyst does not change ΔH ? Explain your choice in one or two sentences.
- 3) In a two-step catalyzed pathway, the first peak is 55 kJ above the reactants and the second peak is 30 kJ above the intermediate. Which step is rate-determining? How can you tell from the diagram?
- 4) Explain why a catalyst can make a reaction practical at lower temperature without changing the amount of energy released per mole of reaction.

Quick self-check (answers): 2) (c) — ΔH depends on reactants/products only. 3) First step — highest peak above its starting level. 4) Lower E_a means more successful collisions at the same T ; ΔH per mole is unchanged.

LESSON 6 — Activation Energy

PART F — Guided Practice + Diploma-Style Questions + Full Worked Solutions

Big Idea

Activation energy is the “start-up cost” of a reaction. Even if a reaction releases energy overall ($\Delta H < 0$), particles still need enough energy at the beginning to reach the activated complex. In this practice set, you will learn to read energy diagrams like a story: where the system starts, what barrier it must climb, where it ends, and what that means for rate and energy flow.

Diploma Focus

Most diploma-style questions do not ask you to memorize definitions—they ask you to interpret diagrams and justify reasoning. Show what each arrow represents (E_a , ΔH) and always state whether you are talking about the system or the surroundings.

Common Error

Mixing up activation energy (E_a) with enthalpy change (ΔH). E_a is the barrier height; ΔH is the difference between start and finish.

STUDENT PRACTICE — Questions Only

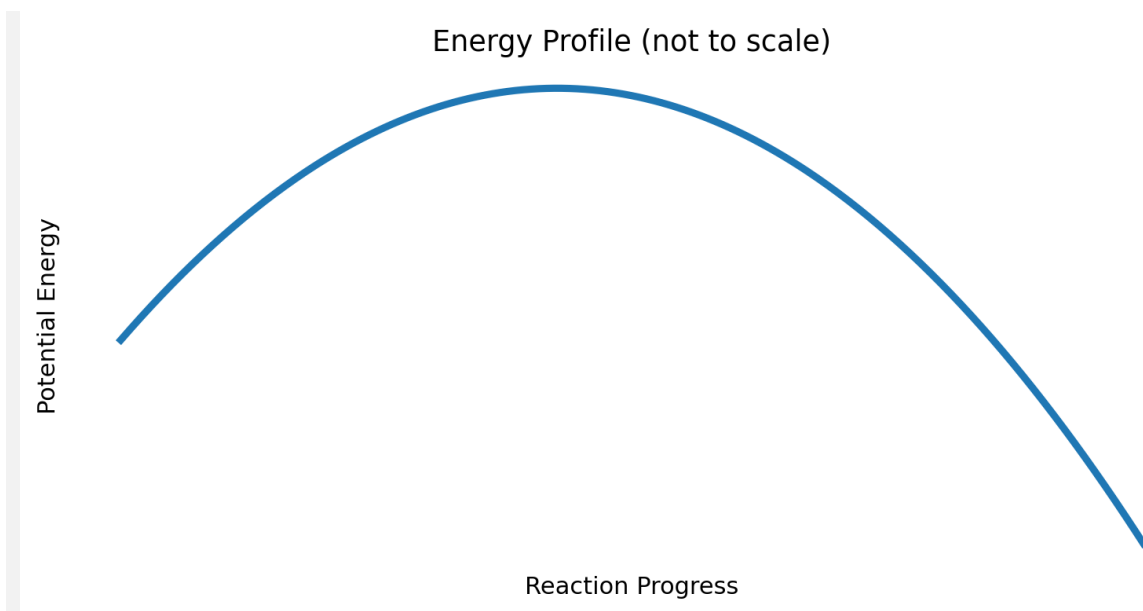
How to Use This Practice Set

Work through the guided questions in order. For written responses, aim for full sentences and clear reasoning. For calculation questions, show the set-up, include units, and state the meaning of the sign (+/-). When you are finished, check your work using the Teacher Version at the end.

GUIDED PRACTICE — Build Your Activation-Energy Skills

Question 1 — Reading an Energy Diagram (E_a and ΔH)

The diagram below shows a reaction energy profile (reaction progress vs potential energy). Assume the reactants start at 80 kJ, the peak is 140 kJ, and the products end at 30 kJ.



- a) In your own words, explain what the peak represents.
- b) Calculate the activation energy for the forward reaction, $E_a(\text{fwd})$.
- c) Calculate ΔH for the reaction and state whether it is exothermic or endothermic.
- d) Describe what happens to the surroundings when this reaction occurs.

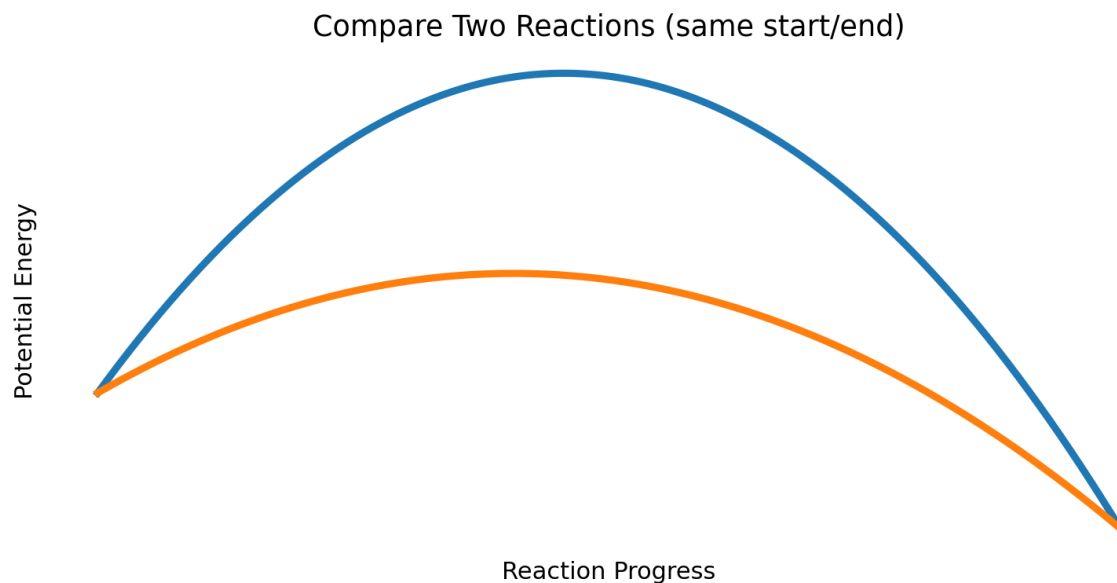
🔧 Question 2 — Forward vs Reverse Activation Energy

Use the same energy profile from Question 1.

- a) Calculate the activation energy for the reverse reaction, $E_a(\text{rev})$.
- b) Explain (in sentences) why $E_a(\text{rev})$ is different from $E_a(\text{fwd})$, even though it is the same reaction running backward.
- c) Write a short rule that connects $E_a(\text{fwd})$, $E_a(\text{rev})$, and ΔH .

🔧 Question 3 — Which Reaction Is Faster?

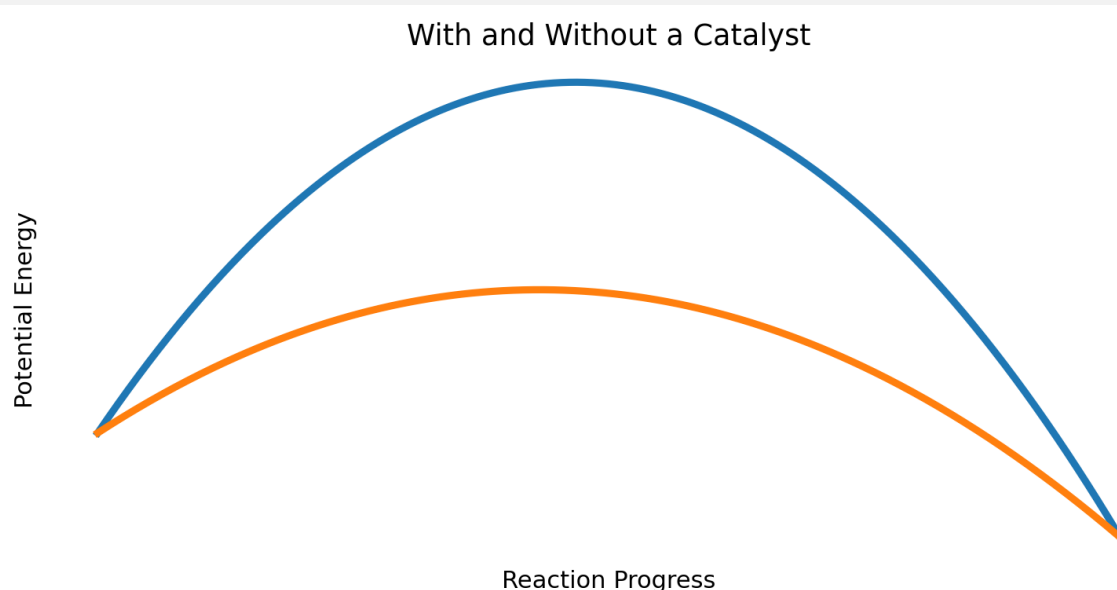
The diagram below shows two possible reaction pathways for the same overall reaction (same reactants and products). One curve has a higher peak (higher E_a), and one has a lower peak (lower E_a).



- a) Without doing any calculations, identify which curve represents the faster reaction at the same temperature.
- b) Explain why—use the idea of an “energy barrier.”
- c) State what stays the same for both curves (hint: think ΔH).

👉 Question 4 — Catalysts on an Energy Diagram

The diagram below shows two energy profiles for the same reaction: one is uncatalyzed, and one is catalyzed.



- a) Which curve represents the catalyzed pathway? How can you tell?
- b) What does a catalyst change on the diagram? What does it NOT change?
- c) In one or two sentences, explain why catalysts improve reaction rate.

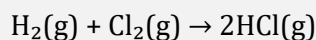
 Question 5 — Correct the Misconception (E_a vs ΔH)

A student says: "If ΔH is very negative, the reaction must happen quickly because it releases lots of energy."

- a) Do you agree or disagree?
- b) Rewrite the student's statement so it becomes scientifically correct.
- c) Use the words activation energy, barrier, and pathway somewhere in your explanation.

 Question 6 — Bond Breaking and Bond Forming (Energy Accounting)

Consider the reaction:



Use the following average bond energies:

$\text{H}-\text{H} = 436 \text{ kJ/mol}$, $\text{Cl}-\text{Cl} = 242 \text{ kJ/mol}$, $\text{H}-\text{Cl} = 431 \text{ kJ/mol}$.

- a) Calculate the energy required to break bonds (reactants).
- b) Calculate the energy released when bonds form (products).
- c) Determine ΔH for the reaction (use: $\Delta H \approx \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$).
- d) Explain how this connects to the idea that 'breaking bonds costs energy' while 'forming bonds releases energy.'

 Question 7 — Temperature Changes Rate Without Changing E_a

You run the same reaction twice, once at 10°C and once at 40°C . The reaction goes noticeably faster at 40°C .

- a) Does the activation energy (E_a) of the reaction change when temperature changes? Explain.
- b) Use particle-motion language to explain why higher temperature increases reaction rate.
- c) State clearly what temperature changes and what it does not change on an energy diagram.

 Question 8 — Real-World Reasoning (Stability vs Reactivity)

Gasoline (a fuel) contains molecules that can release a lot of energy when they burn, yet gasoline can be stored and transported safely.

- a) Use activation energy to explain why gasoline does not burst into flame at room temperature by itself.
- b) Explain what a spark or flame provides to start combustion.
- c) Connect this to why controlling activation energy matters in real-world energy systems.

DIPLOMA-STYLE QUESTIONS — Mixed Response (MC + Written)

 Multiple Choice (circle ONE best answer)

9) A reaction diagram shows products LOWER than reactants. Which statement is correct?

- A. $\Delta H > 0$ and the reaction is endothermic.
- B. $\Delta H < 0$ and the reaction is exothermic.
- C. $E_a < 0$ and the reaction is exothermic.
- D. $E_a > 0$ and the reaction is endothermic.

10) For a reaction, $E_a(\text{fwd}) = 60 \text{ kJ}$ and $\Delta H = -40 \text{ kJ}$. What is $E_a(\text{rev})$?

- A. 20 kJ
- B. 60 kJ
- C. 100 kJ
- D. -100 kJ

11) Which statement about catalysts is BEST?

- A. A catalyst lowers ΔH by making the reaction more exothermic.
- B. A catalyst increases reactant energy so they collide harder.
- C. A catalyst provides an alternative pathway with a lower E_a .
- D. A catalyst is consumed, which releases energy to start the reaction.

12) On an energy diagram, activation energy for the forward reaction is measured as the energy difference between:

- A. reactants and products
- B. products and peak
- C. reactants and peak
- D. reactants and products plus the peak

 Written Response (Diploma-Style)

13) Many students think exothermic reactions should not require any input of energy. Using bond breaking/forming language and the idea of the activated complex, explain why an exothermic reaction still has an activation energy.

14) A two-step reaction has two peaks on its energy profile. Explain how you would identify the rate-determining step from the diagram. Then explain why the overall ΔH depends only on the start and end energies, not on how many steps the reaction takes.

15) STS Connection: Catalysts are used in industry to reduce energy costs and improve safety. Choose ONE example (enzymes in the body, catalytic converters, or industrial ammonia production) and explain how catalysts relate to activation energy, reaction rate, and energy efficiency.

TEACHER VERSION — Full Worked Solutions (with reasoning)

✓ How to Use These Solutions

Use the solutions to check both your answers and your thinking. If your final number matches but your reasoning is unclear, rewrite your explanation using the sample language. Pay special attention to sign conventions (system vs surroundings) and to the difference between E_a (barrier height) and ΔH (start-to-finish change).

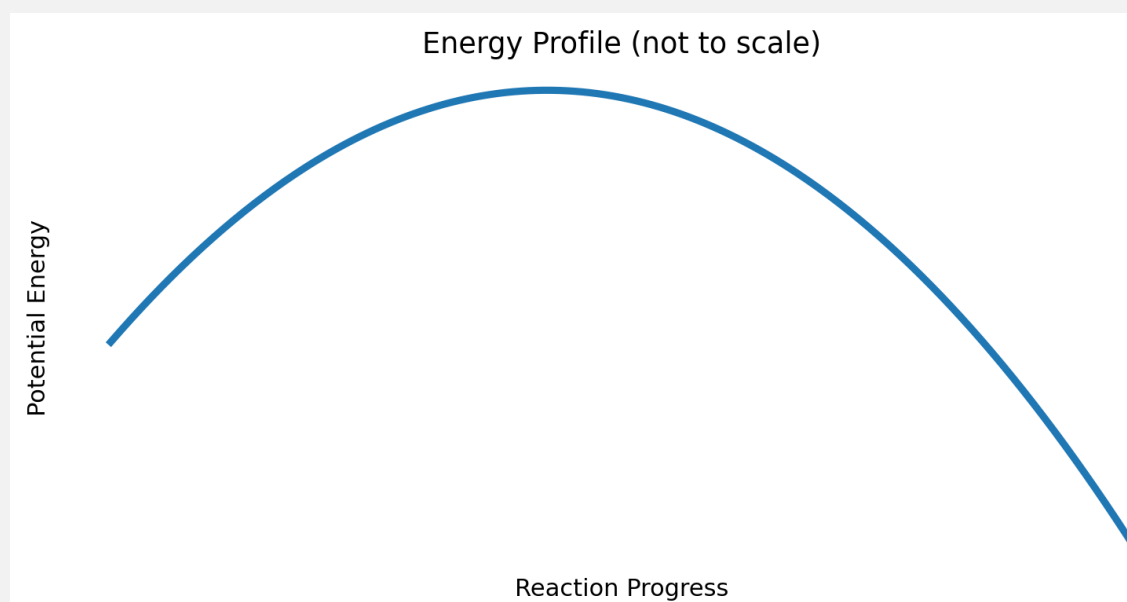
✓ Solution 1

a) The peak represents the activated complex (transition state): the highest-energy arrangement of atoms along the reaction pathway. It is unstable and exists only briefly.

b) $E_a(\text{fwd}) = E(\text{peak}) - E(\text{reactants}) = 140 \text{ kJ} - 80 \text{ kJ} = 60 \text{ kJ}$.

c) $\Delta H = E(\text{products}) - E(\text{reactants}) = 30 \text{ kJ} - 80 \text{ kJ} = -50 \text{ kJ}$. Because ΔH is negative, the reaction is exothermic: the system releases energy overall.

d) If the system releases energy, that energy is transferred to the surroundings. The surroundings warm up (temperature increases) if the energy transfer is mostly as heat.



✓ Solution 2

a) $E_a(\text{rev}) = E(\text{peak}) - E(\text{products}) = 140 \text{ kJ} - 30 \text{ kJ} = 110 \text{ kJ}$.

b) The reverse reaction starts from the products, which are at a lower potential energy

than the reactants in this exothermic case. To reach the same activated complex, the reverse reaction must climb a larger energy gap.

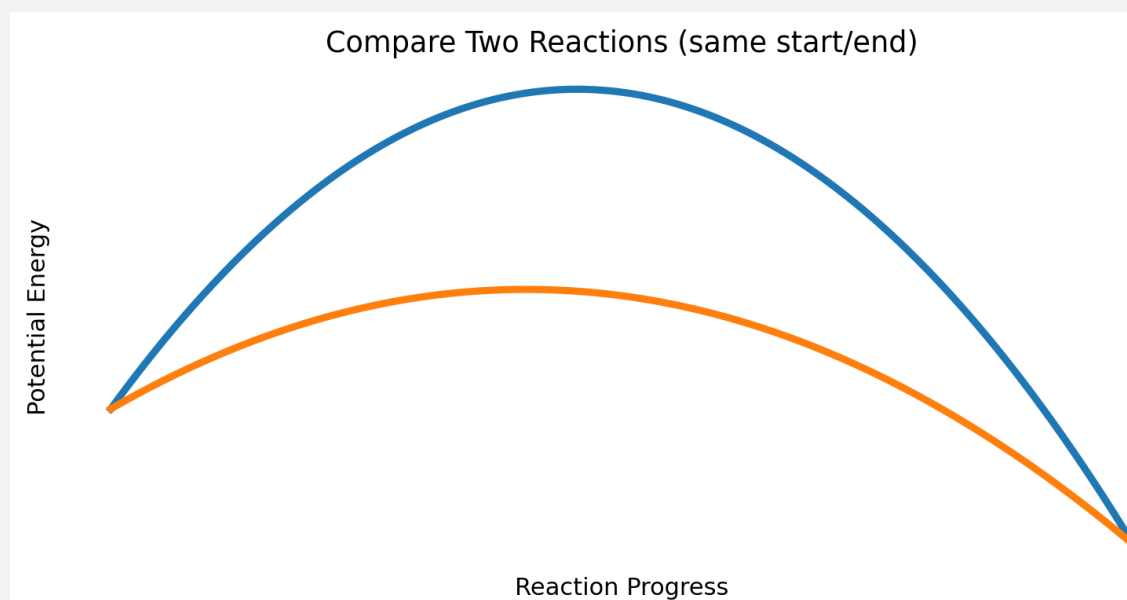
c) Relationship rule: $E_a(\text{rev}) = E_a(\text{fwd}) - \Delta H$. Notice why this works: when ΔH is negative (exothermic), subtracting a negative makes $E_a(\text{rev})$ larger than $E_a(\text{fwd})$.

✓ Solution 3

a) The faster reaction is the one with the LOWER peak (lower E_a). At the same temperature, a greater fraction of particles can surpass a smaller barrier.

b) Activation energy is like a hill the system must climb. A lower hill is easier to climb, so successful collisions happen more often.

c) ΔH stays the same because the reactants and products are unchanged. Only the pathway (and barrier height) differs.

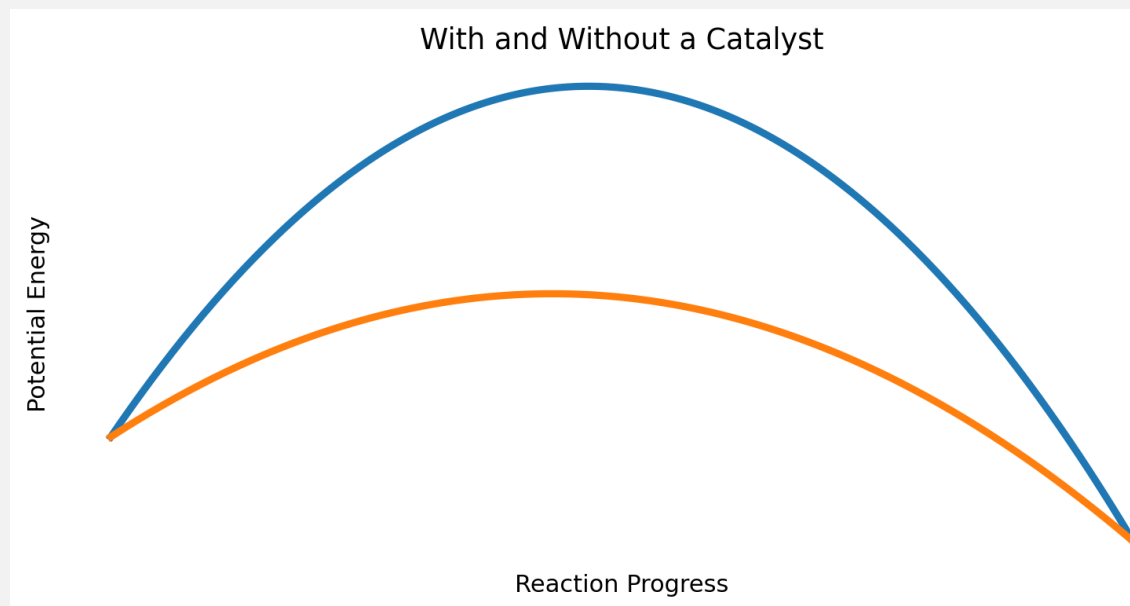


✓ Solution 4

a) The catalyzed pathway is the curve with the LOWER peak. A catalyst lowers the activation energy by providing an alternative pathway.

b) A catalyst changes E_a (peak height). It does NOT change the starting energy (reactants), ending energy (products), or ΔH . ΔH is determined only by initial and final states.

c) With a lower barrier, more collisions have enough energy to reach the activated complex, so the reaction occurs faster.



✓ Solution 5

a) Disagree. A negative ΔH means energy is released overall, but it does not guarantee the reaction happens quickly.

b) Better statement: "A negative ΔH tells us the system ends at a lower energy than it started, but the reaction rate depends on activation energy and the pathway."

c) Key correction: even a very exothermic reaction can be slow if the activation-energy barrier is large.

✓ Solution 6

a) Bonds broken: H—H (436) + Cl—Cl (242) = 678 kJ/mol.

b) Bonds formed: $2 \times \text{H—Cl} = 2 \times 431 = 862$ kJ/mol (released).

c) $\Delta H \approx \Sigma E(\text{broken}) - \Sigma E(\text{formed}) = 678 - 862 = -184$ kJ/mol. Negative means exothermic.

d) Why this works: breaking bonds requires energy input to pull atoms apart; forming bonds releases energy because the product bond is more stable (lower potential energy).

The overall sign depends on whether bond-forming release outweighs bond-breaking cost.

✓ Solution 7

a) E_a is a property of the reaction pathway and does not change just because temperature changes.

b) Higher temperature means particles have higher average kinetic energy and a wider spread of energies. More particles have $E \geq E_a$, so more collisions are successful per second.

c) On an energy diagram, the curve (including E_a and ΔH) stays the same. Temperature changes how many particles can 'get over' the barrier, not the barrier itself.

✓ Solution 8

a) Gasoline combustion is exothermic (large negative ΔH), but it has a significant activation energy. At room temperature, most collisions between gasoline vapour and oxygen do not have enough energy to reach the activated complex.

b) A spark/flame supplies the initial energy needed to overcome the activation barrier—once started, the reaction can sustain itself.

c) In energy systems, controlling activation energy helps balance safety (storage stability) and usefulness (rapid energy release when desired).

✓ Solution 9–12

9) B. Products lower than reactants means $\Delta H < 0$, exothermic.

10) C. Use $E_a(\text{rev}) = E_a(\text{fwd}) - \Delta H = 60 - (-40) = 100 \text{ kJ}$.

11) C. A catalyst provides an alternative pathway with lower E_a .

12) C. $E_a(\text{fwd})$ is measured from reactants up to the peak.

✓ Solution 13

A strong explanation includes these points:

- Bond breaking always requires energy input. Early in the reaction, reactant bonds must

be stretched and partially broken before new bonds can form.

- That bond-breaking 'investment' is the activation energy.
- Once the activated complex is reached, new bonds form and release energy. If the release is greater than the initial cost, the overall ΔH is negative (exothermic).
- So: ΔH describes the start-to-finish change, while E_a describes the barrier you must cross to get the reaction started.

✓ Solution 14

Rate-determining step:

- On a multi-step energy profile, the rate-determining step is the step with the largest activation energy barrier (largest peak relative to the previous valley).
- That step is slowest, so it controls the overall rate.

Overall ΔH :

- ΔH depends only on initial and final energies (reactants vs products). Intermediate steps and peaks change the pathway but do not change the start-to-finish energy difference.
- This is the same 'path-independence' idea you will use again with Hess's Law.

✓ Solution 15

A strong STS response should include:

- A named example (enzymes / catalytic converter / Haber process).
- A statement that catalysts lower E_a by offering an alternative pathway.
- A link to practical outcomes: faster reactions at lower temperature, less energy input, better control, improved safety, reduced emissions or waste.

Example (catalytic converter): It uses metal catalysts to lower E_a for oxidation/reduction reactions that convert CO and NO_x into less harmful gases. Lower E_a means the reactions proceed quickly at exhaust temperatures, improving air quality.

LESSON 6 SELF-CHECK (before you move on)

- ✓ I can explain activation energy as an energy barrier and describe what the peak means on an energy diagram.
- ✓ I can calculate $E_a(\text{fwd})$, $E_a(\text{rev})$, and ΔH when energy values are provided.
- ✓ I can distinguish clearly between E_a (barrier height) and ΔH (start-to-finish energy change).
- ✓ I can explain catalysts as alternative pathways that lower E_a without changing ΔH .
- ✓ I can justify answers using diagram language (reactants, products, activated complex, pathway).

